PIZZA SALES ANALYSIS

Python-Based Sales Data Analysis & Business Insights

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1. Introduction

The Pizza Sales Analysis project focuses on analyzing transactional sales data from a pizza store to identify business performance, customer purchasing patterns, and sales trends.

The project uses Python to process and analyses historical sales data. Pandas is used for data manipulation, aggregation, and calculations, while Matplotlib and Seaborn are used to create visual representations of the analysis.

The main purpose of this project is to transform raw sales data into meaningful business insights that can help management make better decisions regarding sales performance, product demand, inventory planning, and operational activities.

2. Project Objectives

The main objectives of this project are:

- Calculate important sales performance indicators, including total revenue, total pizzas sold, and total orders.
- Calculate Average Order Value (AOV) and average pizzas per order.
- Analyze sales performance by day, hour, and month.
- Identify sales distribution across pizza categories and sizes.
- Identify the best-selling and least-selling pizzas based on revenue.
- Create visualizations to present sales patterns and business insights.

3. Tools and Technologies

The following technologies and libraries were used in this project:

Technology	Purpose
Python	Programming language for data analysis
Pandas	Data loading, cleaning, transformation, grouping, and aggregation
Matplotlib	Creating charts and data visualizations
Seaborn	Creating statistical and categorical visualizations
Jupyter Notebook	Developing and executing the analysis

4. Dataset Description

The project uses a CSV file named `pizza_sales.csv`, containing transactional sales records from a pizza store.

The dataset contains **48,620 records and 12 original columns**.

The important columns used in the analysis are:

Column	Description
pizza_id	Unique identifier for each pizza record
order_id	Identifier for each customer order
pizza_name_id	Unique identifier for the pizza type and size
quantity	Number of pizzas sold
order_date	Date of the order
order_time	Time of the order
unit_price	Price of an individual pizza
total_price	Total transaction value for the pizza record
pizza_size	Size of the pizza
pizza_category	Category of the pizza
pizza_ingredients	Ingredients used in the pizza
pizza_name	Name of the pizza

5. Methodology and Implementation

The project was completed through several stages, starting with data loading and preparation, followed by exploratory data analysis, KPI calculations, and visualization.

5.1 Data Loading

The Pandas library was used to load the CSV dataset into a DataFrame.

The `read_csv()` function was used to import the data, and `head()` was used to inspect the first five records.

This helped understand the dataset structure and identify the available columns.

5.2 Data Inspection and Quality Checking

Initial data inspection was performed using the following Pandas functions:

- isnull().sum() – Checked for missing values in each column.

- duplicated() – Checked for duplicate pizza identifiers.
- info() – Examined the number of records, column names, data types, and memory usage.

Data inspection results:

- Total records: 48,620
- Total original columns: 12
- Missing values: 0 across all columns
- All 48,620 records contained non-null values.

The duplicate check was performed on the `pizza_id` column to identify repeated pizza identifiers.

5.3 Data Type Conversion

The `order_date` and `order_time` columns were initially stored as string values.

The `pd.to_datetime()` function was used to convert both columns into datetime formats.

This conversion allowed date and time information to be extracted and used for further analysis.

5.4 Feature Engineering

Additional analytical columns were created from the existing date and time information.

The following features were extracted:

New Column	Description
order_day	Day of the week when the order was placed
month_num	Numerical representation of the month
order_month	Name of the month
order_hour	Hour when the order was placed

These additional columns made it possible to analyses daily, monthly, and hourly sales patterns.

5.5 Key Performance Indicator (KPI) Calculation

Five important business KPIs were calculated using Pandas.

The formulas used were:

Total Revenue

Sum of all values in the `total_price` column.

Total Pizzas Sold

Sum of all values in the `quantity` column.

Total Orders

Count of unique values in the `order_id` column.

Average Order Value (AOV)

Total Revenue ÷ Total Orders

Average Pizzas per Order

Total Pizzas Sold ÷ Total Orders

These KPIs provide an overview of the store's overall sales performance and customer purchasing behaviors.

5.6 Sales Analysis

The following analytical operations were performed using Pandas `groupby()`, `sum()`, `nunique()`, `agg()`, and `sort_values()` functions.

A. Sales by Day of the Week

Sales revenue was grouped by the day of the week to identify which days generated the highest and lowest revenue.

B. Sales by Hour

Sales revenue was grouped by the order hour to understand customer purchasing patterns throughout the day.

C. Monthly Sales Trends

Monthly revenue and unique order counts were calculated to identify changes in sales performance across the year.

D. Sales by Pizza Category

Revenue was grouped by pizza category, and each category's percentage contribution to total revenue was calculated.

E. Sales by Pizza Size and Category

Sales revenue was grouped by pizza size and category to understand how different pizza combinations contributed to overall revenue.

F. Total Pizzas Sold by Category

The total quantity sold was calculated for each pizza category to understand product demand.

G. Top 5 Best-Selling Pizzas

Pizza revenue was grouped by pizza name and sorted in descending order to identify the five highest-revenue pizzas.

H. Bottom 5 Least-Selling Pizzas

Pizza revenue was sorted in ascending order to identify the five lowest-revenue pizzas.

6. Key Performance Indicator Results

The following results were obtained from the dataset:

KPI	Results
Total Revenue	Rs. 817,860.05
Total Pizzas Sold	49,574
Total Orders	21,350
Average Order Value	Rs. 38.31
Average Pizzas per Order	2.32

These results provide a summary of the store's overall sales performance.

7. Analysis Results, Business Insights and Visualization

7.1 Sales by Day of the Week

The analysis showed the following revenue distribution:

Day	Revenue
Sunday	Rs. 102,116.45
Monday	Rs. 110,471.60
Tuesday	Rs. 115,594.45
Wednesday	Rs. 116,731.20
Thursday	Rs. 121,650.30
Friday	Rs. 129,690.90

Saturday	Rs. 121,605.15

Insight: Friday generated the highest revenue, while Sunday generated the lowest. This information can help management plan staffing and operational activities according to daily sales demand.

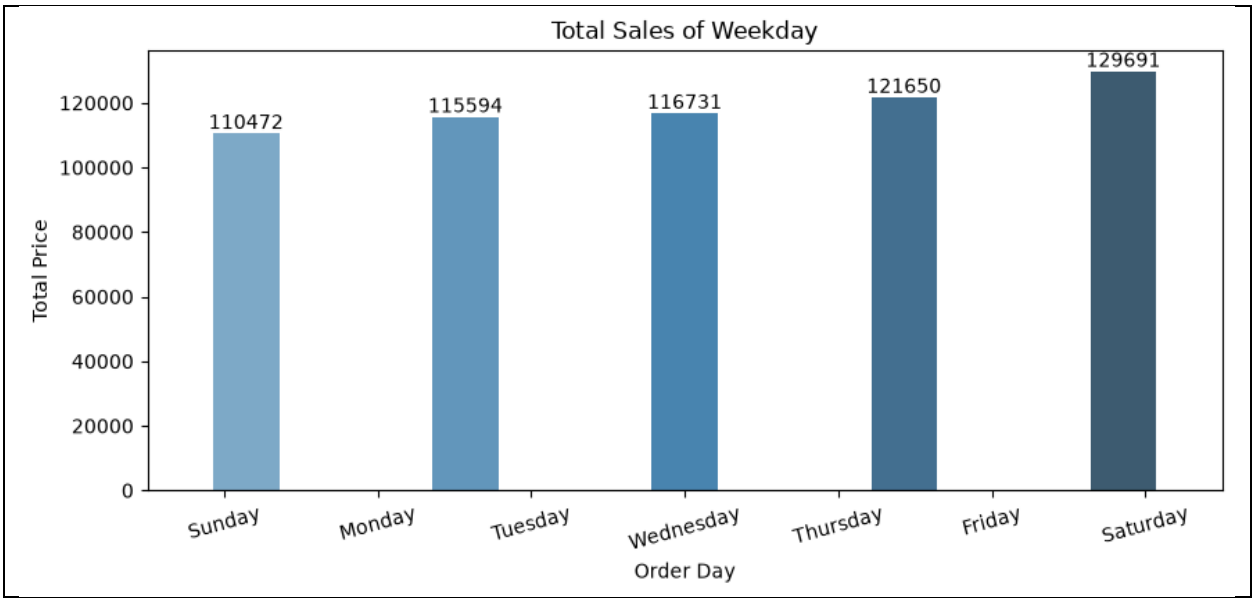


Fig: Sales by Day of the Week – Bar Chart

7.2 Sales by Hour

The hourly analysis showed that:

- 12 PM generated the highest hourly revenue of Rs. 111,877.90.
- 1 PM generated Rs. 106,065.70.
- Sales were also significant during the evening hours.

Insight: The afternoon period, particularly around lunchtime, represents an important sales period. Management can consider appropriate staffing and ingredient preparation during these hours.

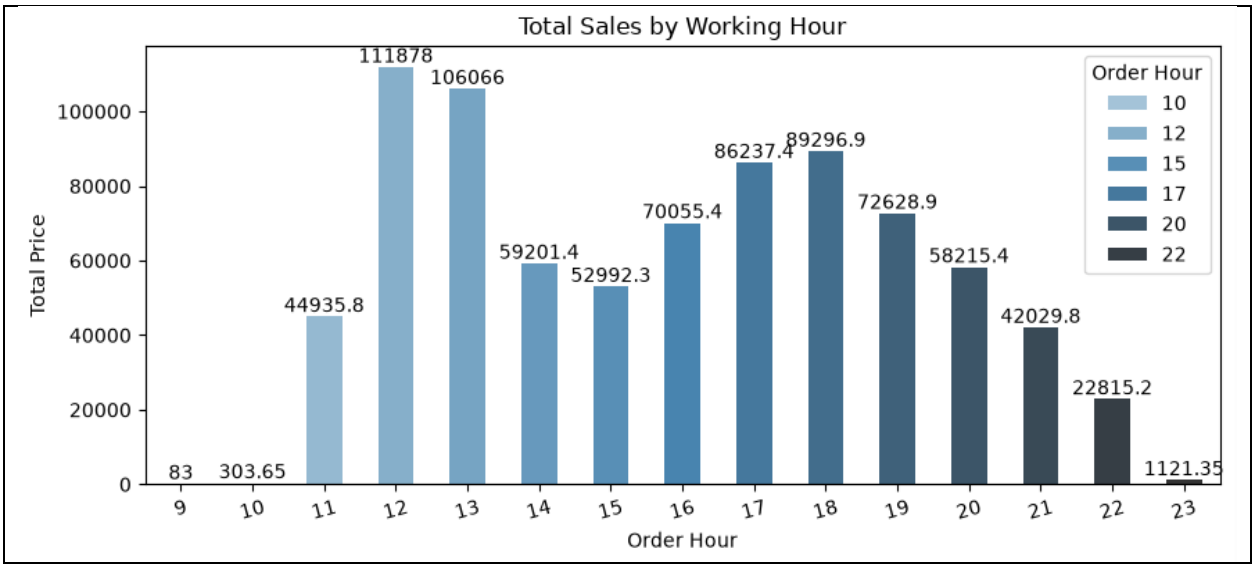


Fig: Sales by Hour of the Day – Bar Chart

7.3 Monthly Sales Trends

Month	Revenue	Orders
January	Rs. 71,620.15	1,929
February	Rs. 64,419.45	1,648
March	Rs. 71,301.40	1,864
April	Rs. 70,312.00	1,829
May	Rs. 67,648.80	1,765
June	Rs. 68,161.45	1,771
July	Rs. 70,880.65	1,860
August	Rs. 69,497.30	1,835
September	Rs. 63,803.70	1,638
October	Rs. 68,152.20	1,782
November	Rs. 71,004.85	1,844
December	Rs. 61,058.10	1,585

Insight: January recorded the highest monthly revenue, while December recorded the lowest. These results help management understand monthly sales variations and plan future promotional activities.

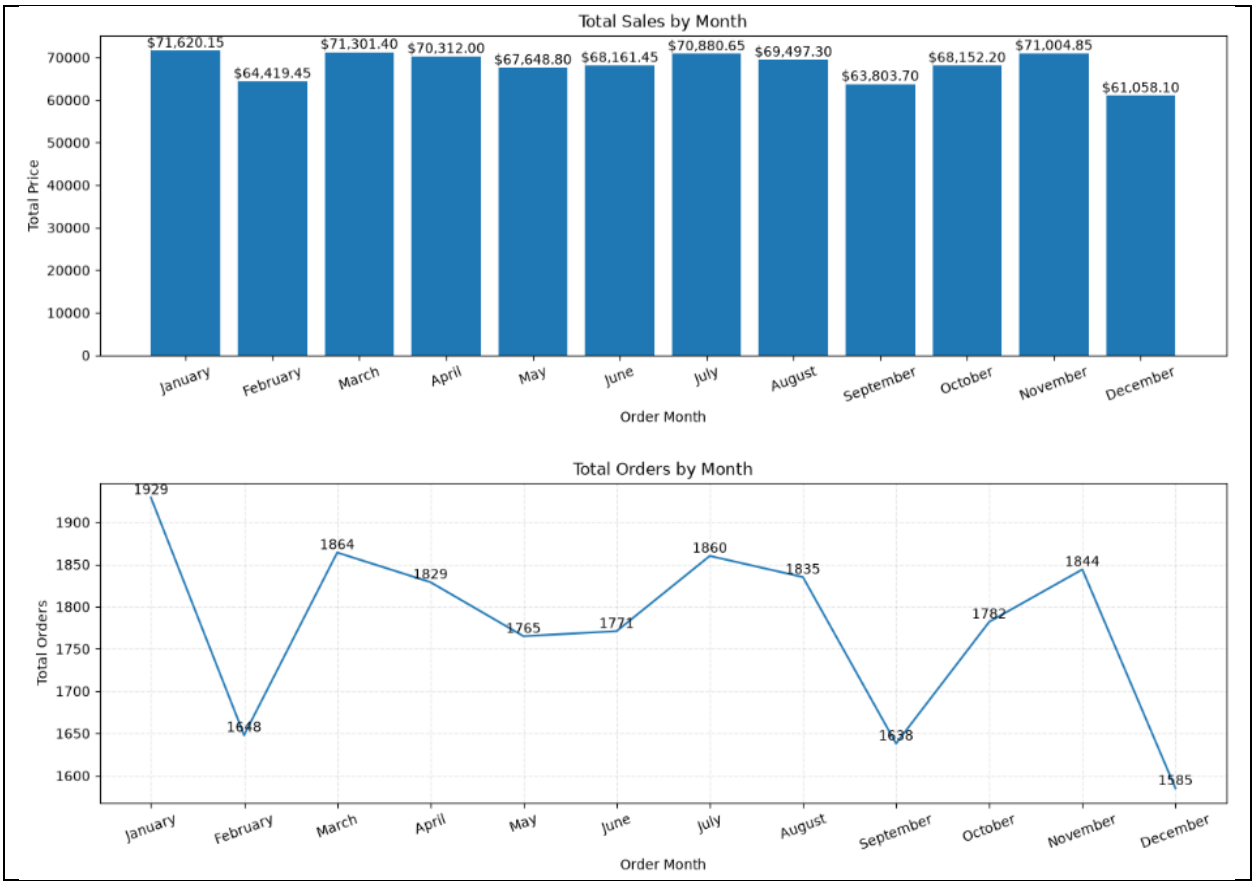


Fig: Monthly Orders – Bar Chart & Line Chart

7.4 Percentage of Sales by Pizza Category

Category	Revenue	Sales Contribution
Classic	Rs. 220,053.10	26.91%
Supreme	Rs. 208,197.00	25.46%
Chicken	Rs. 195,919.50	23.96%
Veggie	Rs. 193,690.45	23.68%

Insight: Classic pizzas contributed the highest percentage of total revenue, while Veggie pizzas contributed the lowest.

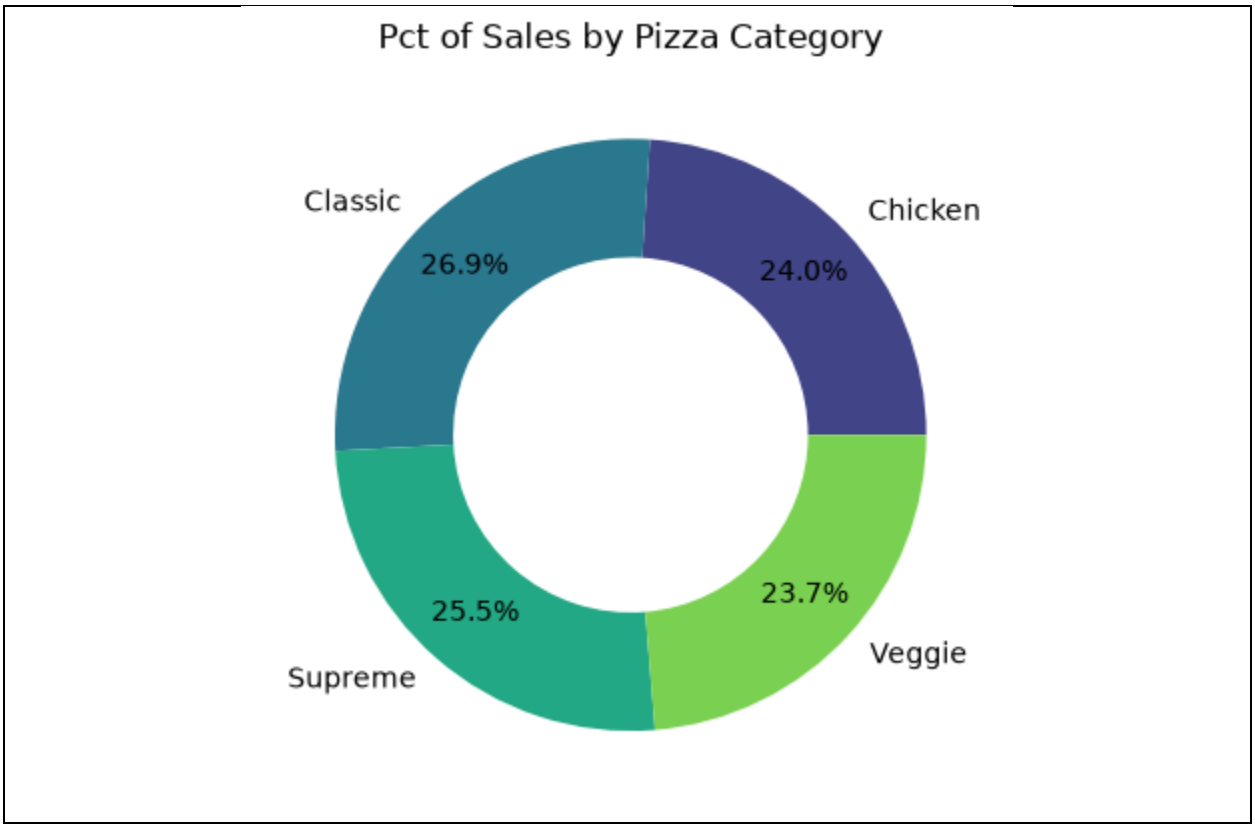


Fig: Percentage of Sales by Category – Doughnut Chart

7.5 Sales Percentage by Pizza Size and Category

The analysis showed that large-sized pizzas contributed a significant portion of total revenue.

The highest individual size-category contribution was:

- Large Veggie: 12.74%
- Large Chicken: 12.51%
- Large Supreme: 11.53%
- Large Classic: 9.11%

Insight: Large-sized pizzas generated strong revenue contributions across all four categories. This information can support product planning and inventory decisions.

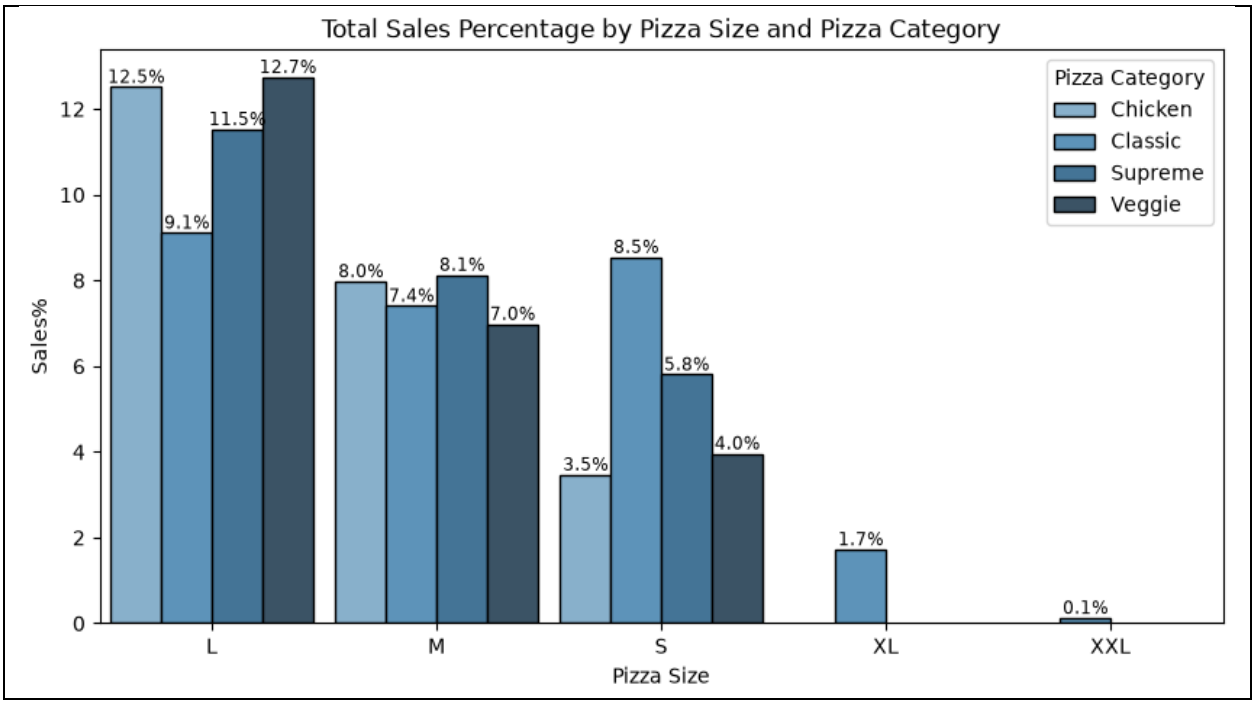


Fig: Sales Percentage by Pizza Size and Category – Bar Chart

7.6 Total Pizzas Sold by Category

Category	Quantity Sold
Classic	14,888
Supreme	11,987
Veggie	11,649
Chicken	11,050

Insight: Classic pizzas had the highest sales quantity, with 14,888 pizzas sold. Chicken pizzas recorded the lowest quantity among the four categories.

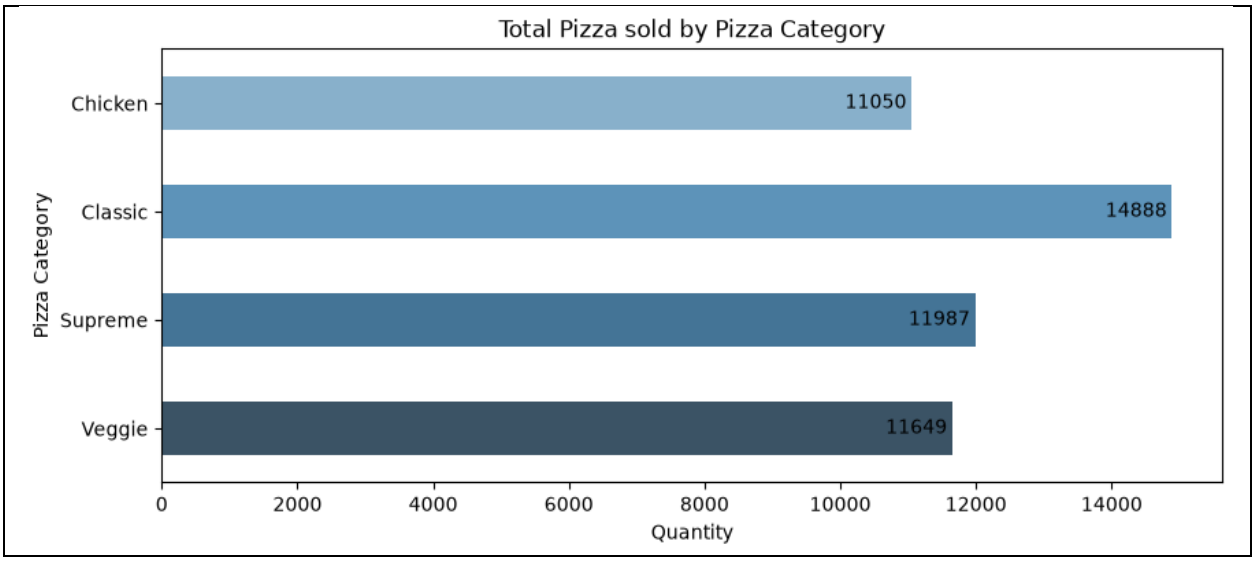


Fig: Total Pizzas Sold by Category – Horizontal Bar Chart

7.7 Top 5 Best-Selling Pizzas by Revenue

Pizza Name	Revenue
The Thai Chicken Pizza	Rs. 43,434.25
The Barbecue Chicken Pizza	Rs. 42,768.00
The California Chicken Pizza	Rs. 41,409.50
The Classic Deluxe Pizza	Rs. 38,180.50
The Spicy Italian Pizza	Rs. 34,831.25

Insight: The Thai Chicken Pizza generated the highest revenue among the analyzed pizza types.

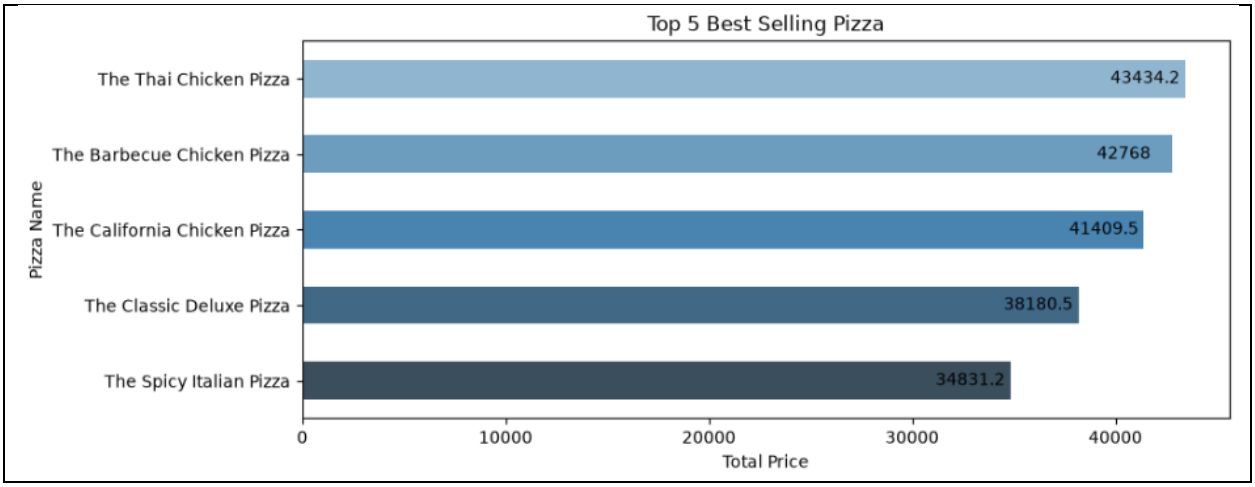


Fig: Top 5 Best-Selling Pizzas – Horizontal Bar Chart

7.8 Bottom 5 Least-Selling Pizzas by Revenue

Pizza Name	Revenue
The Brie Carre Pizza	Rs. 11,588.50
The Green Garden Pizza	Rs. 13,955.75
The Spinach Supreme Pizza	Rs. 15,277.75
The Mediterranean Pizza	Rs. 15,360.50
The Spinach Pesto Pizza	Rs. 15,596.00

Insight: The Brie Carre Pizza generated the lowest revenue among the analyzed pizza types. These products may require further evaluation of customer demand, pricing, and promotional strategies before making menu-related decisions.

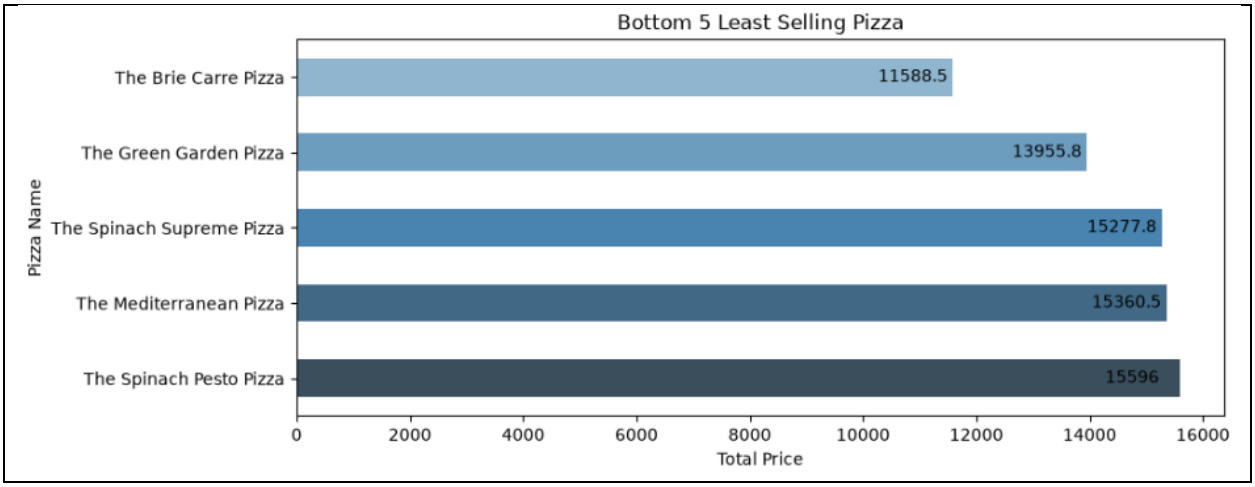


Fig: Bottom 5 Least-Selling Pizzas – Horizontal Bar Chart

8. Business Recommendations

Based on the analysis results, the following recommendations can be considered:

- **Sales Strategy:** Focus promotional activities on high-performing categories and popular pizza types.
- **Inventory Planning:** Maintain appropriate stock levels for ingredients used in frequently purchased pizza categories.

- **Staffing:** Allocate sufficient staff during high-demand periods, particularly Friday and lunchtime.
- **Product Improvement:** Review the performance of least-selling pizzas and evaluate whether improvements or promotional activities are required.
- **Sales Monitoring:** Monitor revenue, order volume, and average order value regularly to understand business performance.

These recommendations are based on the observed sales patterns and should be considered alongside other business factors.

9. Conclusion

The Pizza Sales Analysis project demonstrates the practical use of Python in analyzing transactional sales data and generating meaningful business insights.

Using Pandas, the dataset was inspected, cleaned, transformed, and analyzed to calculate key performance indicators (KPIs) and identify sales trends across different time periods, pizza categories, sizes, and individual products.

Matplotlib and Seaborn were used to create visualizations that present the analytical results clearly. These visualizations helped identify important sales patterns, including the highest-revenue day, peak sales hours, monthly performance, category contributions, and the best- and least-performing pizzas.

Overall, this project provided practical experience in data analysis, data manipulation, exploratory data analysis (EDA), KPI calculation, data visualization, and interpreting analytical results to develop meaningful business recommendations.